Oldham College

Networking Academy

LAB WORKSHEET

ASA Firewall Lab

How a Cisco ASA 5505 decides what to allow: security levels, the default rules, stateful inspection, and using an ACL for exceptions.

Security levels

Every ASA interface has a security level 0–100 — how trusted the network behind it is.

Interface	Level	Meaning
Inside	100	your LAN — most trusted
DMZ	50	public-facing servers
Outside	0	the internet — least trusted

The default rules

- **Higher → lower is allowed** with no rule (Inside→Outside, Inside→DMZ, DMZ→Outside).
- **Lower → higher is denied** by default (Outside→Inside, Outside→DMZ, DMZ→Inside).
- **Stateful inspection:** the ASA tracks connections it allowed out and lets the matching **reply** back in automatically — no rule needed for return traffic.
- **Default-deny inbound:** with no ACL, an unsolicited inbound connection is dropped. "No rule" means deny, not allow.

Making an exception with an ACL

To allow a specific lower→higher flow (e.g. the internet reaching a DMZ web server on 443):

```
access-list OUTSIDE_IN permit tcp any host 203.0.113.10 eq 443
access-group OUTSIDE_IN in interface outside
```

Why a DMZ? A public server sits at level 50, between Outside and Inside. DMZ→Inside is lower→higher, so even a compromised DMZ server cannot start a connection to the trusted Inside LAN.

Tab 2 — Rule practice

Build a rule: read each scenario and build the ACL line with the builder, or type the full command. You must choose the right interface, action, protocol, source, destination and port. **Put rules in order:** drag the lines into the right order, then run the test packets. The ASA stops at the **first match**, and anything unmatched hits the **implicit deny**.

ACE syntax: `access-list NAME extended {permit|deny} {tcp|udp|icmp|ip} SOURCE DEST [eq PORT]`, where SOURCE/DEST = `any` | `host A.B.C.D` | `network mask`. ASA ACLs use **subnet masks** (255.255.255.0), not wildcard masks.

Tab 3 — Configure the ASA

The topology is on the left and the ASA console on the right. Send test packets from any host: the envelope travels to the ASA and gets a PERMIT or DENY badge. If it's allowed, the reply comes back too.

```
enable
configure terminal
interface vlan 10
 nameif inside
 security-level 100
 ip address 192.168.10.1 255.255.255.0
interface vlan 20
 nameif outside
 security-level 0
 ip address 203.0.113.2 255.255.255.248
interface ethernet0/0
 switchport access vlan 20
 no shutdown
interface ethernet0/1
 switchport access vlan 10
 no shutdown
route outside 0.0.0.0 0.0.0.0 203.0.113.1
access-list OUTSIDE_IN extended permit tcp any host 192.168.10.50 eq https
access-group OUTSIDE_IN in interface outside
policy-map global_policy
 class inspection_default
  inspect icmp
write memory
```

- **Guided task:** 12 steps that tick off as you work. ⚡ **Auto-configure** loads the interface, VLAN and route set-up so the class can start straight at the rules (step 6).
- **Independent task:** meet a fictional dental practice's requirements, then press **Run task tests** to produce a test log (expected vs actual) that you can copy.
- Useful checks: `show nameif`, `show interface ip brief`, `show switch vlan`, `show route`, `show access-list` (hit counts), `packet-tracer input outside tcp 198.51.100.20 1025 192.168.10.50 443`.
